Lab: Cost Visibility for Azure Databricks Using Tags

1. Overview

Azure Databricks cost comes from two bills: **DBUs** (Databricks) and **Azure infrastructure** (VMs, disks, networking). Tags are the thread that ties both back to a cost centre, project and owner.

In this lab you define a tag standard, enforce it at every place compute is created, and build cost dashboards in Databricks and Azure Cost Management.

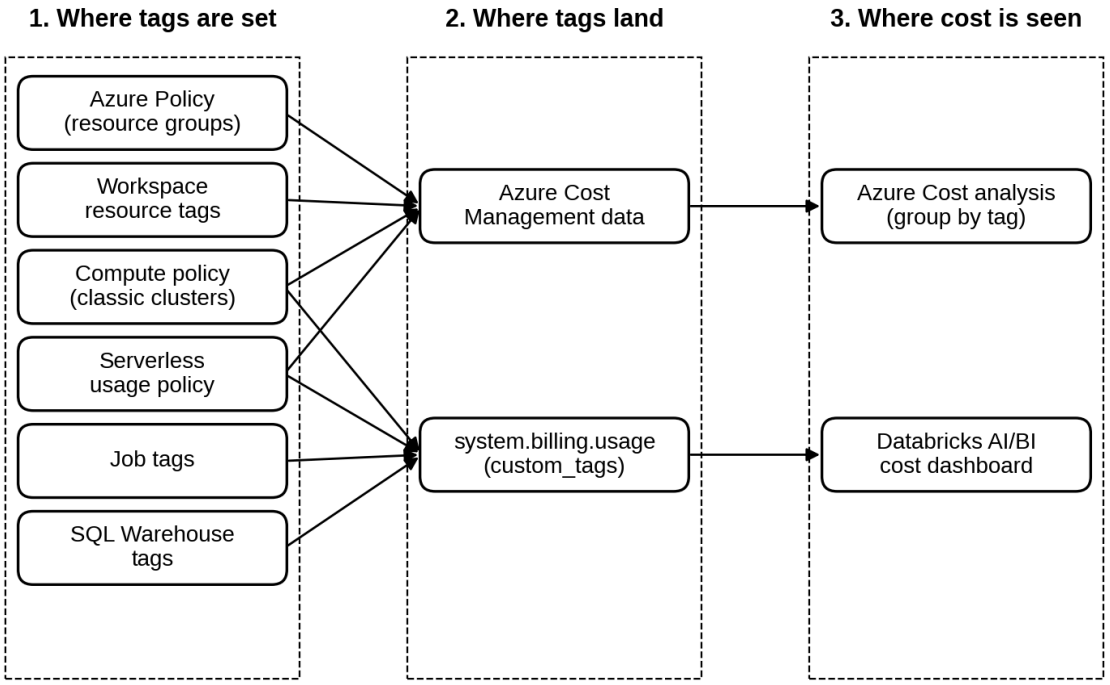


Figure 1. How tags flow from compute to billing dashboards

Tag standard used in this lab (illustrative values). Keys are case-sensitive: spell them exactly as shown.

Key	Value in this lab	Rule
CostCenter	CC-3100 (Actuarial) or CC-3200 (Cat Modelling)	Allowlist
Project	cyclone-pool-analytics	Fixed
Environment	dev	Allowlist: dev, test, prod

Key	Value in this lab	Rule
Owner	Your email address	Must be a valid email

2. Pre-requisite

- Access to the shared Azure Databricks workspace as a **workspace admin**.
- Serverless compute enabled in the workspace.
- Trainer has granted you `SELECT` on `system.billing` and the **Cost Management Reader** role in Azure.
- Replace `<username>` with your short user name and `<your_email>` with your login email throughout.

> **Timing:** billing data is not real-time. Databricks usage appears in system tables within a few hours; Azure Cost Management can take 8 to 24 hours. Complete Parts A and B today and Part C in the next session.

3. Learning Objectives

- Explain where Azure Databricks cost comes from and where tags are applied.
- Enforce tags with Azure Policy, compute policies, job tags, SQL Warehouse tags and serverless usage policies.
- Query tagged usage from `system.billing.usage` and build a cost dashboard.
- Build a tag-based cost view in Azure Cost Management.
- Measure untagged spend as a compliance KPI.

4. Step-by-step guide

Part A: Azure tagging and Azure Policy (trainer demo)

The trainer performs these steps in the Azure portal. Follow along and note where each setting lives.

Step A1: Tag the resource group and the workspace

1. In the Azure portal, open **Resource groups** and select the resource group that holds the Databricks workspace.

2. Click **Tags**, add the four keys from the tag standard, then click **Apply**.
3. Open the **Azure Databricks** workspace resource, click **Tags**, add the same four keys and click **Apply**.
4. Open the workspace **Overview** and click the **Managed Resource Group** link. This is where Databricks creates cluster VMs and disks.

> Workspace tags are applied to resources Databricks creates in the managed resource group. That group is locked by Databricks, so Azure Policy cannot edit tags there. Databricks applies them itself.

Step A2: Require tags on resource groups

1. Open **Policy**, then **Definitions**. Search for **Require a tag on resource groups**.
2. Click **Assign policy**. Set **Scope** to the training subscription.
3. On the **Parameters** tab, set **Tag Name** to CostCenter. Click **Review + create**, then **Create**.
4. Repeat for Project, Environment and Owner.
5. Test it: try to create a new resource group with no tags. Azure rejects it with a policy error.

> New assignments can take up to 30 minutes to take effect.

Step A3: Inherit tags from the resource group

1. In **Definitions**, search for **Inherit a tag from the resource group if missing**.
2. Click **Assign policy**. Set **Scope** to the workspace resource group.
3. On **Parameters**, set **Tag Name** to CostCenter.
4. On **Remediation**, tick **Create a remediation task** and keep the **system assigned managed identity**. Click **Review + create**, then **Create**.
5. Repeat for the other three keys. Existing untagged resources in the group (storage, Key Vault, networking) now pick up the tags.

Step A4: Turn on tag inheritance in Cost Management (optional)

1. Open **Cost Management**, then **Settings** and **Manage subscription**.
2. Select **Tag inheritance**, click **Edit**, enable it and click **Save**.

> This copies subscription and resource group tags onto cost records, so costs are grouped correctly even for resources that lack their own tags. It is available on EA and MCA billing accounts.

Part B: Enforce tags in Databricks

Step B1: Create a compute policy

1. In the workspace sidebar, click **Compute**, open the **Policies** tab and click **Create policy**.
2. Name it ct-<username> and add a short description.
3. Under **Definitions**, open **Advanced options** and click **Edit definition as JSON**.
4. Paste the definition below, then click **Create**.

```
{
  "custom_tags.CostCenter": {
    "type": "allowlist",
    "values": ["CC-3100", "CC-3200"]
  },
  "custom_tags.Project": {
    "type": "fixed",
    "value": "cyclone-pool-analytics"
  },
  "custom_tags.Environment": {
    "type": "allowlist",
    "values": ["dev", "test", "prod"],
    "defaultValue": "dev"
  },
  "custom_tags.Owner": {
    "type": "regex",
    "pattern": "^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\\.[A-Za-z]{2,}$"
  },
  "autotermination_minutes": {
    "type": "range",
    "maxValue": 30,
    "defaultValue": 20
  },
  "num_workers": {
    "type": "range",
    "maxValue": 2,
    "defaultValue": 1
  }
}
```

Rule type	Effect
fixed	Value is set by the policy. Users cannot change it.
allowlist	Users must pick from the listed values.
regex	Users must type a value that matches the pattern.

Step B2: Test the policy

1. Click **Compute**, then **Create compute**. Set **Policy** to ct-<username>.
2. Leave **Owner** empty and click **Create**. Creation is blocked by a policy error.
3. Set **Owner** to <your_email> and **CostCenter** to CC-3100. Click **Create**.
4. When it is running, open the compute and check its **Tags**. You will see your four tags plus the default tags added by Databricks: Vendor, Creator, ClusterName and ClusterId.

Step B3: Create a tagged job

1. Create a notebook named cost_lab_job, set the language to **SQL** and paste the query below.

```
SELECT pickup_zip, COUNT(*) AS trips, ROUND(AVG(fare_amount), 2) AS avg_fare
FROM samples.nyctaxi.trips
GROUP BY pickup_zip
ORDER BY trips DESC;
```

2. Click **Jobs & Pipelines**, then **Create** and **Job**. Name it cost-lab-<username>.
3. Add a **Notebook** task pointing to cost_lab_job.
4. For **Compute**, add a new **job cluster**, select policy ct-<username> and fill in **Owner** and **CostCenter**.
5. In the **Job details** panel, click **Tags**, then add Pipeline = cost-lab-<username>.
6. Click **Run now** and wait for the run to succeed.

> Job tags are added to the job cluster, so costs can be split below Project level, per pipeline.

Step B4: Create a tagged serverless SQL Warehouse

1. Click **SQL Warehouses**, then **Create SQL warehouse**. Name it wh-<username>.
2. Set **Cluster size** to 2X-Small, **Auto stop** to 10 minutes, **Scaling** to min 1 and max 1, and **Type** to **Serverless**.

3. Under Advanced options, add the four tag keys with your values. Click Create.
4. Open the SQL Editor, select wh-<username> and run the query from Step B3.

Step B5: Create a serverless usage policy

Serverless notebooks, jobs and pipelines do not use compute policies. Serverless usage policies (formerly budget policies) carry their tags instead.

1. Click your user name in the top bar, then Settings. Under Workspace admin, click Compute.
2. Next to Serverless usage policies, click Manage, then Create.
3. Name it sup-<username>, add the four tag keys with your values and click Create.
4. Open the policy's Permissions and give yourself the User permission.
5. Create a new notebook, attach it to Serverless, open the Environment side panel and select sup-<username> as the usage policy.
6. Run the query from Step B3 in this notebook.

> A user who is assigned only one usage policy gets it applied automatically. Admins may see several and must pick one.

Part C: Build billing dashboards (next session)

Step C1: Confirm your tags reached the billing table

1. Open the SQL Editor on wh-<username> and run the query below.
2. Confirm you see rows for ALL_PURPOSE, JOBS, SQL and serverless INTERACTIVE usage, each with your tags in custom_tags.

```
SELECT billing_origin_product,
    sku_name,
    custom_tags,
    SUM(usage_quantity) AS dbus
FROM system.billing.usage
WHERE usage_date >= current_date() - 2
  AND custom_tags['Owner'] = '<your_email>'
GROUP BY ALL;
```

Step C2: Build a Databricks cost dashboard

1. Click Dashboards, then Create dashboard. Name it Cost by Tag - <username>.

2. Open the Data tab, click Create from SQL and paste the query below. Rename the dataset to cost_by_tag.

```
SELECT u.usage_date,
       u.billing_origin_product                    AS product,
       u.sku_name,
       COALESCE(u.custom_tags['CostCenter'],  'untagged') AS cost_center,
       COALESCE(u.custom_tags['Project'],     'untagged') AS project,
       COALESCE(u.custom_tags['Environment'], 'untagged') AS environment,
       COALESCE(u.custom_tags['Owner'],       'untagged') AS owner,
       SUM(u.usage_quantity)                       AS dbus,
       SUM(u.usage_quantity * p.pricing.default)        AS list_cost_usd
FROM system.billing.usage u
JOIN system.billing.list_prices p
  ON  u.sku_name = p.sku_name
  AND u.cloud    = p.cloud
  AND u.usage_start_time >= p.price_start_time
  AND (p.price_end_time IS NULL OR u.usage_start_time < p.price_end_time)
WHERE u.usage_date >= current_date() - 30
GROUP BY ALL;
```

3. Add a second dataset named untagged with the query below.

```
SELECT ROUND(100 * SUM(IF(u.custom_tags['Project'] IS NULL,
                          u.usage_quantity * p.pricing.default, 0))
             / SUM(u.usage_quantity * p.pricing.default), 1) AS untagged_pct
FROM system.billing.usage u
JOIN system.billing.list_prices p
  ON  u.sku_name = p.sku_name
  AND u.cloud    = p.cloud
  AND u.usage_start_time >= p.price_start_time
  AND (p.price_end_time IS NULL OR u.usage_start_time < p.price_end_time)
WHERE u.usage_date >= current_date() - 30;
```

4. On the Canvas, add the visuals in the table below.
5. Add two Filter widgets on owner and environment, then click Publish.

Visual	Dataset	Settings
Counter	cost_by_tag	Value: SUM(list_cost_usd)
Bar	cost_by_tag	X: project, Y: SUM(list_cost_usd), Colour: product
Bar	cost_by_tag	X: owner, Y: SUM(list_cost_usd)

Visual	Dataset	Settings
Line	cost_by_tag	X: usage_date, Y: SUM(list_cost_usd), Colour: product
Counter	untagged	Value: untagged_pct (target: under 5%)

> Costs use list prices in USD. ARPC's negotiated rate may differ; apply the discount in the dashboard if needed.

Step C3: Build a tag view in Azure Cost Management

1. In the Azure portal, open **Cost Management**, then **Cost analysis**. Set the scope to the training subscription.
2. Set **Group by** to **Tag**, then **CostCenter**. Set the date range to **Last 7 days** and granularity to **Daily**.
3. Click **Add filter**, choose **Tag**, then Project = cyclone-pool-analytics.
4. Change **Group by** to **Service name**. You will see **Azure Databricks** (DBUs) and **Virtual Machines** (infrastructure) as separate lines.
5. Click **Save as** and name the view ARPC - Databricks cost by tag.

> Serverless compute has no VM line in Azure. Its infrastructure is included in the DBU rate, so the DBU line is its full cost.

Part D: Clean up

1. Terminate your all-purpose compute.
2. Stop or delete wh-<username>.
3. Keep ct-<username> and sup-<username> for later labs.

5. Outcome

- Resource groups and the workspace carry the four standard tags, enforced by Azure Policy.
- Classic compute cannot start without valid tags, because of your compute policy.
- Jobs, SQL Warehouses and serverless usage are all tagged.
- A published Databricks dashboard shows cost by project, owner, product and day, plus untagged spend %.
- A saved Azure Cost Management view shows DBU and infrastructure cost by tag.